CSC240_BST/QueType.h

```
// Header file for Queue ADT.
#ifndef QUE_H
#define QUE_H
#include <new>
#include <cstddef>
class FullQueue
{};

class EmptyQueue
{};
typedef char ItemType;

struct NodeType;

class QueType
{
public:
    QueType();
    // Class constructor.
    // Because there is a default constructor, the precondition
    // that the queue has been initialized is omitted.
    QueType(int max);
    // Parameterized class constructor.
    ~QueType();
    // Class destructor.
    QueType(const QueType& anotherQue);
    // Copy constructor
    void MakeEmpty();
    // Function: Initializes the queue to an empty state.
    // Post: Queue is empty.
    bool IsEmpty() const;
    // Function: Determines whether the queue is empty.
    // Post: Function value = (queue is empty)
    bool IsFull() const;
    // Function: Determines whether the queue is full.
    // Post: Function value = (queue is full)
    void Enqueue(ItemType newItem);
    // Function: Adds newItem to the rear of the queue.
    // Post: If (queue is full) FullQueue exception is thrown
    //       else newItem is at rear of queue.
    void Dequeue(ItemType& item);
    // Function: Removes front item from the queue and returns it in item.
    // Post: If (queue is empty) EmptyQueue exception is thrown
    //       and item is undefined
    //       else front element has been removed from queue and
    //       item is a copy of removed element.
private:
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48
49    NodeType* front;
50    NodeType* rear;
51  };
52  #endif
53
```